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Intuition and Reasoning: A Dual-Process Perspective

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A lay definition of intuition holds that it involves immediate apprehension in the absence of reasoning. From a more technical point of view, I argue also that intuition should be seen as the contrastive of reasoning, corresponding roughly to the distinction between Type 1 (intuitive) and Type 2 (reflective) processes in contemporary dual process theories of thinking. From this perspective, we already know a great deal about intuition: It is quick, provides feelings of confidence, can reflect large amounts of information processing, and is most likely to provide accurate judgments when based on relevant experiential learning. Unlike reasoning, intuition is low effort and does not compete for central working memory resources. It provides default responses which may—or often may not—be intervened upon with high effort, reflective reasoning. Intuition has, however, been blamed for a range of cognitive biases in the psychological literatures on reasoning and decision making. The evidence indicates that with novel and abstract problems, not easily linked to previous experience, intervention with effortful reasoning is often required to avoid such biases. Hence, although it seems that intuition dominates reasoning most of the time—both in the laboratory and the real world—it can indeed be a false friend.

> **Intuition** . . . Immediate apprehension by the mind without the intervention of reasoning. — *Oxford English Dictionary*

The lay definition of *intuition* takes us some of the way toward finding the technical meaning needed for cognitive psychological study. Intuition is indeed in some sense *immediate* in that it is not preceded by any conscious process of reflective thinking. I find it most interesting that the *Oxford English Dictionary* defines intuition in part by the *absence* of something – a process of reasoning. A contrast between two such kinds of thinking has been made repeatedly by philosophers and psychologists for centuries, often writing in ignorance of the relevant work of others (Frankish & Evans, 2009). Many terms are used, but in essence the idea is that one kind of thinking is slow and reflective, whereas the other is quick and intuitive. Sometimes the conscious nature of the former is emphasised, but intuitions are not actually unconscious, as they often manifest as feelings. What is apparently not conscious about them, however, is any experience of where they come from and how they are justified.

Like the *Oxford English Dictionary*, I believe we can only hope to find a useful understanding of intuition by considering it in combination with its contrastive: reasoning. I use the term *reasoning* here in a restrictive sense of slow, reflective, deliberative and goal-oriented thinking. I avoid the term *conscious* in this definition for reasons that become clear later. In terms of modern psychology, two kinds of thinking corresponds with a family of *dual-process* theories that have sprung up across many fields of psychology, including the study of learning, memory, attention, social cognition, thinking, reasoning, and decision making (e.g., Epstein, 1994; Evans & Over, 1996; Reber, 1993; Sloman, 1996; Smith & DeCoster, 2000; Stanovich, 1999). A recent review of a number of these theories and associated empirical research is provided by Evans (2008). In the tradition observed by Frankish and Evans (2009), however, many of these theories have been developed independently and without cross-reference. Nevertheless, they all draw upon a broadly similar distinction between two types of mental process, which we can summarize as follows:

Type 1 process: fast, intuitive, high capacity
Type 2 process: slow, reflective, low capacity

Even with this minimal level of description, we see that we have now a technical addition to the lay meaning of the term *intuition*. Compared with Type 2 processes, Type 1 thinking can handle larger amounts of information or process it in parallel. Intuition and the closely related notion of *insight* have similarities with perception, as many authors have noted. It is both conscious and unconscious in the same sense. For example, when we perceive a face or listen to a piece of music, we have a conscious experience that includes a representation of the object and accompanying emotional

experiences. We have, however, no conscious experience of the complex information processing that precedes this experience. Indeed, we could not have such experience because the parallel and rapid processing that is required far exceeds the capacity of working memory, which seems to be the only part of the mind that is consciously accessible, at least in a cognitive sense. Intuition similarly has a product, which may be a judgement, action, or feeling of confidence, but again without awareness of process.

I use the term *working memory* here in the sense of a central and singular resource of limited capacity closely linked with executive and attentional functions. A working memory system of this kind is both implied by the large research program inspired by Baddeley and Hitch (1974; Baddeley, 2007) and the welter of findings linking individual differences in working memory capacity with a host of cognitive functions (Barrett, Tugade, & Engle, 2004). (Working memory capacity is measured by the number of items that can be kept in short-term storage while performing a competing cognitive task.) Recently, dual-process researchers have been linking their definition of Type 2 processing to the involvement of this working memory system (Evans, 2008) and a number of experimental manipulations intended to demonstrate dual processes either measure or manipulate working memory capacity, as I discuss next. So a tentative definition of intuition might link it with cognitive processes that do not require access to working memory. This is why it is fast, high capacity, and lacking consciousness of the underlying process. But if we are to look to dual-process theory for an answer to the problem of intuition, we must first clarify some potential confusion in the current literature.

Dual Processes and Modes: Is Intuition a Matter of Cognition or Personality?

Recently (Evans, 2009, 2010) I have discussed a number of problems that have arisen in the interpretation of research on dual-processes. One issue is that true dual-process theories, which distinguish between two *types* of cognitive processing, can be confused with those that simply describe two *modes* of processing, for example, the holistic and analytic thinking styles that have been proposed to distinguish Eastern and Western cultures (Buchel & Norenzayan, 2009; Nisbett, Peng, Choi, & Norenzayan, 2001). Modes differ from types in several important ways. Whereas types of thinking are attributed to separate underlying cognitive mechanisms, modes may simply reflect the operation of a *single* mechanism in different styles. Modes of thinking can lie on a continuum and be malleable; they can relate to personality, culture, motivation, and situation. The type hypothesis is much stronger: Type 1 and 2 processes are discrete and fundamentally different. Thus researchers predict that cognitive ability relates to Type 2 but not Type 1 processing, that working memory loads or short time constraints will selectively affect Type 2 processing, and so on. Modes may be culture and personality dependent, but types cannot be. In the large individual differences programme of Keith Stanovich (1999, 2010; Stanovich & West, 2000) it has emerged that type differences are related to cognitive ability, whereas modes of thinking are related to personality and cognitive style measures, such as the extensively researched Need for Cognition scale (Cacioppo & Petty, 1982). Rational action, as Stanovich (2009c, 2010) has recently argued, may need a combination of the two: It requires not only sufficient cognitive capacity but a disposition to approach problems with a rational thinking style.

Now we may also speak of intuition as a personality trait, when we describe a person as "intuitive." What does this mean, and how could this be reconciled with a dual process theory at the type level? I have argued recently that a number of dual-process theories of reasoning and decision making have a structure that I describe as *default-interventionist* (e.g., Evans, 1989, 2006b; Kahneman & Frederick, 2002; Stanovich, 1999, 2010). Such theories assume a rapid Type 1 (intuitive) process provides a quick default solution to a problem, which may be either accepted or intervened upon with explicit Type 2 reasoning. When such intervention occurs, the default intuition may (or may not) be overridden. There is now neural imaging evidence to support this proposal. The brain appears to detect conflict when intuitive and reflective processes would deliver different judgments—in the *anterior cingulate cortex*—and when the default response is overridden, in the *right lateral prefrontal cortex* (De Neys, Vartanian, & Goel, 2008; Goel & Dolan, 2003; Tsujii & Watanabee, 2009). There is also evidence from the social neuroscience literature that distinct neural systems are responsible for implicit and explicit social judgments and that the former are inhibited when the latter take control (Lieberman, 2007, 2009).

A question of contemporary interest in dual-process research is what determines the likelihood of intervention with Type 2 processing (Evans, 2009). Possibilities include the amount of time available, presence or absence of competing demands, motivation to think "rationally," and feelings of confidence in the initial intuition (Thompson, 2009). To this list we may certainly add dispositional thinking styles, related to personality. On this basis, a person who is "intuitive" might be unkindly described as lacking Need for Cognition or being deficient in rational thinking disposition (Stanovich, 2009c). This is because they rely on their initial feeling rather than checking it out with explicit reasoning. But this negative spin presupposes that Type 2 thinking is in some sense *superior* to Type 1 processing. I would suggest that authors making this

assumption have generally been focused on experimental reasoning and decision tasks that are abstract and novel in nature, requiring Type 2 reasoning. Other authors have claimed that implicit processing may be normal and effective when problems are complex in nature (Berry & Dienes, 1993; Reber, 1993) or when people already have relevant experience and expertise (Klein, 1999; Reyna, 2004). More controversially, some have claimed that reliance on "gut feelings" and unconscious processing is generally advantageous (Dijksterhuis, Bos, Nordgren, & von Baaren, 2006; Gigerenzer, 2007; Gladwell, 2005).

Frederick (2005) devised the Cognitive Reflection Test comprised of just three deceptively simple questions. Each had an intuitively suggested wrong answer and a correct answer that could be quite easily found with a little reflective reasoning. An example is, "If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? ____ minutes." This phrasing of this problem primes the intuitive error of "100 minutes," whereas the correct answer (5 min) requires only the simplest numerical reasoning. In spite of this, people of high IQ are very prone to make the intuitive error on these problems. For example, Harvard undergraduates average only 1.43 of 3 correct, and only 20% managed to solve all three. Moreover, Frederick discovered that intuitive errors were related to impulsive personality traits. People getting these wrong were, for example, more likely to say they would pay Amazon for next-day delivery rather than wait a couple days for free shipping.

The Cognitive Reflection Test is one of many tasks used in the cognitive literature on reasoning and decision making that indicates that intuition can be false friend. Intuitive responses *feel* right, but that does not mean that they are right. The relation between impulsivity and reliance on intuition is also interesting and appears in a quite different literature—the psychology of gambling. Gambling is highly popular in Western societies despite the fact that almost all gamblers suffer long-term financial losses. Literally millions of people fly into Las Vegas alone every month to try their luck, mostly at chance games like fruit machines and roulette where the odds are rigged in favor of the house. Moreover, an alarming 1 to 2% of the population become pathological gamblers, a problem that has been linked with both impulsivity and attention deficit disorder (Raylu & Oci, 2002). The psychology of gambling is complex, because it is possible that the thrill of engagement compensates for financial losses to an extent, making it less irrational than first appears. However, it seems likely that false intuitions about chance and probability play a part. I return to this topic later.

I am suggesting here that intuition is *both* a matter of cognition and personality. Reliance on immediate feelings of rightness or confidence, essentially means trusting your Type 1 processes, without intervention by slower more effortful Type 2 reasoning. Many factors affect the likelihood of such intervention, but one of these certainly seems to be personality type, with a negative link to impulsivity and a positive link to various measures of rational thinking dispositions. However, authors appear strongly divided on the question of whether reliance on intuition is good for thinking. Although Gigerenzer (2007) advised us to rely on our "gut feelings" in decision making, Stanovich (2009b, 2009c) argued that failure to apply reflective reasoning is major cause of bias and irrationality in the real world. Of course, the truth may lie in between. For example, there are many famous reports of intuition and sudden insight playing a role in mathematics (e.g., Poincaré, 1927), but no one would accept anything less than the most detailed and explicit reasoning as an actual proof. It is also more than likely that intuitive thinkers will do better in some fields of activity and analytic thinkers in others.

Intuition and the Two Minds Hypothesis

As already indicated, dual-process *type* theories require the assumption that Type 1 and 2 processes are rooted in distinct cognitive mechanisms. A number of authors have been tempted to talk of *two systems* for learning, reasoning, and social processing (Epstein, 1994; Evans & Over, 1996; Reber, 1993; Sloman, 1996), which take responsibility for Type 1 and 2 processing, respectively. Stanovich (1999) dubbed these System 1 and System 2. These terms have become popular and applied recently in the field of judgement and decision making by Kahneman and Frederick (2002). However, two system theories have been criticized by those unsympathetic to dual-process theory (e.g., Gigerenzer & Regier, 1996; Keren & Schul, 2009; Osman, 2004; Shanks & St John, 1994) and also recently questioned and radically revised by those working within the paradigm (Evans, 2006a, 2010; Stanovich, 2009a; 2010).

Although some dual-system theorists have confined themselves to a particular domain such as learning or reasoning, others have speculated much more widely. System 1 has hence been associated with a form of cognition that is evolutionarily ancient, linked with animal cognition and operating mostly unconsciously, whereas System 2 has been described as recent, distinctively human, and conscious (see Evans, 2008, for a detailed review of different dual system theories). Used in this way, however, the term *system* is most unsatisfactory. Could System 1, for example, really be a single system responsible for implicit learning, intuitive judgement, and implicit social cognition? Is there a System 2 that does deductive reasoning, planning, consequential decision making, explicit learning, and a whole host of other cognitive functions known to be

related to working memory capacity? Certainly there is no neuroscientific evidence to support such simplistic claims. In fact, imaging studies suggest that there are multiple cognitive systems involved even in different deductive reasoning tasks (Goel, 2008).

Given the lack of coherence in the two systems approach, authors critical of dual-process theories may argue that we should abandon them in favor of a single-process approach, or confine ourselves to cognitive models of individual tasks. However, I have suggested instead that we should move from taking about two systems to the proposal of two *minds*. The two minds theory (Evans, 2010) inherits a number of characteristics of earlier dual system accounts. Hence, there is an old (intuitive) mind that evolved early and is strongly linked with animal cognition and a new (reflective) mind that evolved recently and is distinctively human. However, the two minds theory is not simply a rebranding of System 1 and 2 as Mind 1 and 2 as sceptical readers might suspect. First, each mind is composed of multiple systems. In developing this idea, I am indebted to the philosopher Samuels (2009), who suggested that it would make more sense to talk of Type 1 and 2 systems rather than Systems 1 and 2. In other words, what people have been calling System 1 and 2 really correspond to two *families* of systems that have the Type 1 and 2 characteristics.

In the two minds theory, each mind has access to different and distinct forms of knowledge together with corresponding learning and memory systems. Type 1 systems within the old mind utilize knowledge gained from experiential learning, using evolutionarily ancient mechanisms, which then manifest as intuitive judgments and behavioral dispositions as well as any instinctively programmed behaviors (much clearer in nonhuman animals). By contrast, Type 2 systems in the new mind manipulate explicit representations through working memory. These are generated by underlying mechanisms such as the ventral visual system (conscious perceptions), episodic memory, and high-level outputs of linguistic and pragmatic processing (explicit meaning). Of course, in terms of their internal working, these new mind support systems are *Type 1*, in that they generally have a modular nature with inaccessible dedicated processes, which are hardwired into the brain. Hence, they are fast, high capacity, consciously inaccessible, and so on. The key difference is that old mind systems (e.g., habit learning, skill learning) are *autonomous*, meaning they can control behavior directly, whereas the Type 1 systems in the new mind first must cooperate with working memory.

Each Type 2 system can be thought of as an ad hoc committee whose membership is chosen to have just the expertise required for the task at hand and that is disbanded on its completion. Each such system will be temporarily formed and will have access to whatever modular support systems are required. For example, someone engaged in a reasoning task will require use of vision, language, and pragmatic systems to create relevant explicit representations. But the proposal that working memory must also be engaged is sufficient to give Type 2 systems their defining characteristics: Only one can function at a time, and each is limited in speed and processing capacity and correlated in its efficacy with individual differences in cognitive capacity. The new mind hence draws upon different knowledge systems from the old mind: Those which provide explicit representations into working memory. Not only are the internal workings of such representational systems clearly Type 1, but the mechanisms responsible for allocation of working memory resources (formation of the current Type 2 system) must also, perforce, be controlled by unconscious mechanisms. The feeling that we have of being consciously in control of our behavior is very largely an illusion (Bargh, 2005; Wegner, 2002).

In the two minds theory, the new or reflective mind is not strictly seen as *uniquely* human because the basis for working memory, attention, and recruitment of higher level control processes are all present in other higher animals (Toates, 2006). However, it is seen as uniquely developed in humans due to the coevolution of language, meta-representation, and a greatly enlarged prefrontal cortex (Evans, 2010). The two minds are also associated with two very distinct forms of rationality. Both are instrumental—serving to achieve goals—but both the goals pursued and mechanisms for achieving them differ. Old mind rationality depends on the pursuit of short-term goals, satisfying an immediate need, by mechanisms which depend upon past successes. These have been programmed by experiential learning (e.g., instrumental conditioning) or by evolution, which in both cases encourage repetition of behaviours that were successful in an earlier environment. By contrast, new mind rationality is driven by anticipation of the future in the form of consequential decision making. This requires an entirely different mechanism, the ability to "decouple" suppositions from actual beliefs and run mental simulations (see also Stanovich, 2010). Thus *hypothetical thinking* is the key facility provided by the new mind that distinguishes our cognition from that of other animals (Evans, 2007; Evans & Over, 1996). Hypothetical thinking (as well as language) is required for all distinctively human activities, such as the invention of mathematics; the ability to engage in engineering design; and the capacity to write, comprehend, and enjoy novels and dramas. It also follows that meta-representations (the representation of mental representations) is a key requirement for the new mind for the following reason. You only need mental representation to believe something, but you need meta-representation to *suppose* something. And such suppositions must be represented explicitly, so that they are distinguished from actual beliefs.

In application to reasoning, decision making, and social judgment I call the old mind "intuitive" and the new mind "reflective" (Evans, 2010). Autonomous Type 1 systems operating through the old mind have the potential to control behavior directly and frequently do so, as when I make an habitual drive to work with my reflective mind freed up to think about the day's work ahead. However, the new mind always has the potential to intervene, as happens when a hazardous traffic situation is suddenly encountered. In the case of judgment and decision making, we call a behavioral propensity emanating from the old mind an *intuitive* judgement. If we rely on our intuition, "we" (our brains) do what feels right without reflecting on it or allocating any precious working memory resources. However, these intuitions do post conscious feelings, which enable the new mind to intervene on default intuitions just as it may on default driving behavior. Whether or not we actually engage reflective reasoning depends upon a whole host of factors. These are best illustrated by examining some of the literature on dual processing.

Intuition in Reasoning Tasks

Some psychologists working in the field of deductive reasoning appear to assume that their enterprise is one of studying the process of explicit reasoning: the manipulation of rules and facts to derive conclusions. I am not and have never been one of them. I started my first text book on the field with these words:

> In one sense this book is only about deductive reasoning. In another sense it is about language comprehension, mental imagery, learning processes, memory organization and the nature of human thought. The first sense is defined by the *paradigms* employed; the second by the natures of the psychological *processes* which the paradigms evoke. (Evans, 1982, p. 1)

Although it is true that those engaged in studying the psychology of reasoning are generally interested in explicit reasoning processes, it is also evident, as I demonstrated in this early review, that that many different kinds of cognitive processes are brought to bear on performance of the tasks that are used in an attempt to study reasoning. And it turns out that many of these are implicit or intuitive in nature. So the title of this section is not as paradoxical as it may seem: The attempt to study reasoning has actually led us to learn a lot about intuition.

The deduction paradigm proper involves presenting people with logical arguments comprised of one or more premises and a conclusion that may or may not follow (a variant involves asking participants to generate their own conclusions). It is conventional to use participants who have had no formal training in logic and to instruct them (a) to assume the truth of the premises and (b) to endorse (or generate) only conclusions which *necessarily* follow from the premises. The historical origin of this rather strange practice is clear: a strong philosophical and psychological tradition that assumes that people must have some in-built capacity for logical reasoning to be rational thinkers (see Evans, 2002). Thus the paradigm tests people's ability to reasoning logically without having been trained to do so. An intensive use of the paradigm between around 1960 and 2000 led to the following findings: (a) People are very poor at solving such reasoning tasks, (b) they exhibit many cognitive biases, and (c) reasoning is highly content dependent. Although early researchers, like Peter Wason, inferred that people were irrational, the longer term view developed that logicism itself was mistaken: Logic cannot be the basis for rationality in an uncertain world (Evans, 2002; Evans & Over, 1996; Oaksford & Chater, 1998; Oaksford & Chater, 2007).

Working in this field myself since the 1970s, I was less interested in arguments about how deduction might be achieved (e.g., rules vs. mental models) and more interested in understanding exactly what people were doing on these tasks. A key publication was that of Evans, Barston, and Pollard (1983), often cited as the foundation for the modern study of belief bias in reasoning. In fact, belief bias was first demonstrated by Wilkins (1928), but our article established the modern paradigm and the main findings. Participants were given classical syllogisms comprising two premises and a conclusion, relating three terms. They were given full deductive reasoning instructions (assume the premises, draw necessary conclusions), and the syllogisms used fell into four categories depending upon whether the conclusion followed necessarily from the premises (valid/invalid) and whether the conclusion presented was believable or unbelievable. Examples are shown in Table 1, together, with the observed rates at which people endorsed their conclusions.

Three main findings observed by Evans et al. (1983) have been replicated in many later studies (e.g., Klauer, Musch, & Naumer, 2000) and can be seen in Table 1. First, people endorse more valid than invalid conclusions, as you would expect if they were reasoning logically. Second, and contrary to the instructions given, they endorse more believable than unbelievable conclusions: the belief bias effect. Third, the belief bias is stronger on invalid syllogisms, leading to a significant belief by logic interaction. Evans et al. showed that the logic-belief conflict was *within* participants and the same person might resolve it different ways on different trials. This was one of the key findings in the development of dual-process accounts of reasoning in which the validity effect was attributed to Type 2 reasoning and the belief bias effect to a Type 1 intuition. However, more recent analyses (Evans, Handley, & Harper, 2001; Klauer et al., 2000; see also

Table 1. *Examples of the Four Types of Syllogism Used by Evans, Barston, and Pollard (1983) Together With Acceptance Rates Combined Over Three Experiments.*

Valid-believable	
No police dogs are vicious	
Some highly trained dogs are vicious	
Therefore, some highly trained dogs are not police dogs	89%
Valid-unbelievable	
No nutritional things are inexpensive	
Some vitamin tablets are inexpensive	
Therefore, some vitamin tablets are not nutritional	56%
Invalid-believable	
No addictive things are inexpensive	
Some cigarettes are inexpensive	
Therefore, some addictive things are not cigarettes	71%
Invalid-unbelievable	
No millionaires are hard workers	
Some rich people are hard workers	
Therefore, some millionaires are not rich people	10%

Note. From *Hypothetical Thinking: Dual Processes in Reasoning and Judgement* (p. 88) by J. St. B. T. Evans, 2007, Hove, UK: Psychology Press. Copyright 2007 by Psychology Press. Reprinted with permission.

Oakhill, Johnson-Laird, & Garnham, 1989) have suggested that there are actually two different belief biases in this task: (a) a Type 1 belief bias consisting of an across the board tendency to say yes to believable and no to unbelievable conclusions, and (b) a Type 2 belief bias which motivates reasoning selectively to prove believable or disprove unbelievable conclusions. With the standard paradigm, the Type 2 bias produces the observed interaction. This is because on invalid-believable problems, people can find a model of the premises that support the conclusion and hence can maintain it; on valid-unbelievable problems, however, any attempt to search for a model of the premises that excludes the conclusion will fail, as none can be found.

Is there direct evidence that the Type 1 belief bias is intuitive? There are several findings that suggest that it is. Recall that one definition of intuition is that it is immediate. Evans and Curtis-Holmes (2005) used a speeded task method in which participants were required to respond in less than half the time they would normally take. This altered the usual pattern of results in three ways: (a) The validity effect was greatly reduced, (b) the belief bias effect was increased, and (c) the Belief $\times$ Logic interaction disappeared. All three findings are consistent with the hypothesis that the speeded task enforces reliance on intuition, thus enhancing the Type 1 belief bias and inhibiting both logical and motivated (Type 2) reasoning. Similar findings have been reported by De Neys (2006) when participants were asked to reason with a concurrent working memory load. He also found better performance on belief–logic conflict problems for participants of higher working memory capacity.

In a recent study, Shynkarkuk and Thompson (2006) directly investigated the role of intuition in syllogistic reasoning. Participants were instructed to give two answers: the first that came to mind within a short time limit and a second given after a free period of reflection. Participants also rated their subjective feelings of confidence in their answers. Participants were more confident in their second answers, after a period of reflection, but not more accurate. In fact, there was no relation at all between confidence and logical accuracy in this study. This contrasts with studies of meta-cognition and memory, in which confidence ratings normally have a positive correlation with accuracy (Koriat, 1993). Moreover, people were more confident in their conclusions about belief-laden materials, which we know to causes belief biases, than in syllogisms with abstract content. These findings suggest that intuition plays a strong role in reasoning experiments, but an unhelpful one.

In a recent study, Morsanyi and Handley (2010) presented syllogistic problems and simply asked participants to rate how much they *liked* their conclusions. The results were remarkably similar to those observed on the standard task with the usual main effects and interaction. This applied whether or not participants were instructed respond intuitively on the basis of their feelings, or on the basis of how a person reasoning logically would respond. What is striking here is that people like valid conclusions as well as believable ones. Is validity in some sense intuitive also? That it may be is suggested by recent findings that physiological or neurological indicators may signal conflict detection even when the belief bias dominates (De Neys, Moyens, & Vansteenwegen, 2010; see also De Neys et al., 2008, for a related finding on a different task). This is a theoretical puzzle, because you cannot know there is a conflict without knowing whether the argument is valid, something generally assumed to require Type 2 reasoning. However, it has also been suggested that syllogisms provide heuristic cues that are highly correlated with validity (Chater & Oaksford, 2004).

Peter Wason, the founder of the modern psychology of reasoning, demonstrated how cognitive biases in reasoning could be intuitively compelling. In the "2 4 6" task (Wason, 1960), participants are asked to work out a rule that the experimenter has in mind which classifies "triples" of three whole numbers. They are given an example that conforms to the rule—2 4 6—and asked to generate triples of their own: In each case the experimenter tells them whether the generated triples conform or not. Typically, people test hypotheses like "ascending with equal intervals" and generate positive examples: 10 20 30; 1 2 3; 34 36 38; and so on, in each case receiving positive feedback. But when they announce the rule they are told they are wrong and have to carry on. The actual rule is any ascending sequence. Participants get very frustrated and often go

on generating similar triples offering alternative verbal formulations of the same rule (Wason, 1968; see also Evans, 2007; Poletiek, 2001, for recent reviews and discussion of work with this task). They find it hard to think of a negative test, such as 1 2 4, which would disprove their rule.

Wason (1966) also devised a reasoning problem, known as the four card selection task, which was intensively studied for the next 30 years or so. In this case participants are given a conditional statement such as, "If there is an A on one side of the card then there is a 3 on the other side of the card," and asked to try to discover whether it is true or false. They are told it applies to cards that have a letter on one side and a number on the other and are shown four such cards, with exposed faces that show A, D, 3, 7. They are instructed to turn over those cards and *only* those cards that could prove whether the rule is true or false. Most people choose A and 3 (sometimes just A), although the correct answer according to standard logic is A and 7. They must try to find a card that has an A on one side and does *not* have a 3 on the other, which would falsify the statement. Only the A and 7 (not a 3) could do this.

Wason's original explanation was a verification bias, which implies faulty Type 2 reasoning. However, he retracted this when I demonstrated a *matching bias* on the task. If the rule is changed to "If there is an A on one side of the card then there is NOT a 3 on the other side of the card," participants still choose A and 3, the cards matching the values in the statement, and they are now logically correct (Evans & Lynch, 1973; for a review of much subsequent research, see Evans, 1998). For whatever reason, the matching cards are intuitively compelling, suggesting that card choices are under control of Type 1 processes. Participants will come up with rationalizations of these choices in the context of the instructions whether or not the negation is present (Lucas & Ball, 2005; Wason & Evans, 1975), arguing that they are verifying or falsifying the statement as applies in either case. Later research showed that people focus their attention selectively on matching cards (Ball, Lucas, Miles, & Gale, 2003; Evans, 1996). Only a small minority of people with very high general intelligence appear to be able to overcome matching bias and attend to the 7 card on the standard problem (Stanovich & West, 1998). However, recent analyses suggest that participants do reason about the cards to which they attend, often leading them to withhold choice of an irrelevant card, even when it matches (Evans & Ball, 2010).

When the selection task uses realistic materials and is placed in a familiar context, it is often much easier to solve. For example, if you are told to play the role of a police officer checking that the following rule is being obeyed, "If people are drinking alcohol then they must be over 18 years of age," the four cards can be used to represent four drinkers. One side shows the beverage and the other side the age of the drinker. Given the cards "beer," "Coke," "21 years of age," and "16 years of age," most people will choose to investigate the beer drinker and the *underage* drinker, which is logically correct and not subject to matching bias (Griggs & Cox, 1982). Does this mean that people have switched from faulty intuition to good logical reasoning? Not so. The evidence suggests that they have switched from faulty intuition to *good* intuition, as their experience of permission and obligation rules provides helpful cues. On such rules, attention is directed from the start toward the correct cards (Evans, 1996) and there is little relation between these choices and measures of cognitive ability (Stanovich & West, 1998). It has even been argued that solving such problems requires no reasoning at all (Sperber & Girotto, 2002).

This brief survey of the psychology of reasoning allows us to draw a most interesting conclusion. The performance of tasks specifically designed to look for explicit Type 2 reasoning are often strongly influenced, or even dominated by, Type 1 intuitive processes. This finding is quite remarkable when you consider that (a) these tasks routinely include instructions designed to engage effortful and systematic reasoning and (b) most employ university students of above population average intelligence as their participants. It compels us to the conclusion that intuition rather than reasoning is the dominant form of human thinking. Although such intuitions can lead to correct answers, they more often generate cognitive biases as well as inspiring feelings of confidence in the wrong answers.

Why does reasoning apparently play so little part in reasoning experiments? One view is that reasoning evolved for the purposes of argumentation (Mercier & Sperber, in press) and hence is more often used to construct justifications for intuitive responses than to generate the answers themselves. However, even if reasoning did evolve in that way it may have been "exapted" for another purpose. It seems impossible to explain the unique intellectual, technological, and cultural achievements of the human species without assuming that genuine Type 2 thinking can underlie our problem solving and decision making. Stanovich (2009c) certainly believes so but has suggested that people are by nature "cognitive misers" who minimize the deployment of effortful reasoning, hence often relying on intuition. Although this may often be successful where we have relevant experiential learning, the reasoning literature shows that it can also have potentially disastrous consequences for people required to reason in a modern technological world. Stanovich stressed that the *inclination* to reason is related to rational thinking disposition rather than general intelligence, which instead predicts the accuracy of any reasoning that gets applied. This suggests the importance of education and training in persuading people, where appropriate, to reason explicitly rather than to rely on intuition.

Intuition in Judgment and Decision Making

Everyday experience suggests that decisions can be made both intuitively and reflectively. For example, if a friend asks you what you would like to drink, the answer will often just appear instantly as what you feel like on that occasion. Of course, you may also engage in reflection: You may feel like a gin and tonic but settle for an orange juice as you will be driving your car shortly. This example seems similar to the notion of Type 2 reasoning overriding a default Type 1 intuition in the reasoning research previously discussed. It also illustrates what I termed new mind rationality earlier in this article: Long-term goals and anticipated consequences may take over from immediate satisfaction of desires. Such overrides in everyday decision making occur frequently: the student with a hangover who wants to stay in bed but gets up to attend a lecture, the married person who resists a tempting sexual opportunity, the dieter who refuses a favorite food, and so on. Intuitive decision making need not arise from immediate desires, however, and may instead reflect lazy thinking or cognitive biases as in many of the examples already discussed in the reasoning field. As the earlier cited work of Frederick (2005) illustrates, people well capable of applying some basic reasoning to a problem may nevertheless rely on immediate and faulty intuitions.

The theory of rational decision making, which was imported into psychology from the discipline of economics, requires that a rational person should anticipate the consequences of their decisions, estimating the probability and utility of various outcomes, combining the two to calculate the *expected utility* of each action, and then choosing the action that maximizes this quantity (von Neumann & Morgenstern, 1944; introduced to psychology by Edwards, 1954, 1961). In psychological terms, this requires sophisticated hypothetical thinking: The individual must carry out mental simulations of the future consequences of all possible actions. For many years, psychological research on decision making was focused on demonstrating violations of decision theory rather than elucidating the underlying cognitive processes (for reviews of relevant literature, see D. J. Koehler & Harvey, 2004). The famous prospect theory article by Kahneman and Tversky (Kahneman & Tversky, 1979; see also Tversky & Kahneman, 1992) demonstrated numerous violations of expected utility as well as proposing an alternative descriptive account of risky choice.

One reason that people may fail to maximize expected utility is that they are not actually thinking consequentially, that is, people do not engage in sufficient reasoning about the consequences of their choices. More cognitively oriented laboratory research has demonstrated this directly (Baron, 1994), whereas studies of expert decision making in the field also found little evidence of classically rational decision making (Klein, 1999). However, in the latter case it is often observed that the correct course of action comes to mind immediately for an expert decision maker. Where relevant experiential learning has taken place, intuition may be a reliable basis for decision making in many cases. Klein (1999) also observed, however, that when problems are novel or unusual, explicit mental simulations may be carried out. I have argued with many examples (Evans, 2007) that when people do conduct such mental simulations they tend to focus on a single hypothesis, unless there is good reason to give it up. This explains many phenomena, including the perseverance with a faulty hypothesis on the Wason 2 4 6 problem, discussed earlier. I call this the "singularity principle" of hypothetical thinking.

In the reasoning research discussed earlier, I showed that even when people are reasoning, their focus may be restricted by preconscious intuitive processes. For example, people reason about the cards to which they attend on the selection task, but attention is cued by intuitive processes like matching bias. They may apply reasoning to try to prove a believable conclusion or to disprove an unbelievable one, but not vice versa. In the same way, there are many examples of restricted attention in decision making tasks. Legrenzi, Girotto and Johnson-Laird (1993) noted that when asked to decide whether they would spend an evening in a particular way, people do not request to know what alternatives are available. Instead, they focus on the case given, for example, whether to attend an expensive but high-quality opera, an apparent case of the singularity principle. Many studies have also shown that the way in which the experimenter *frames* a decision problem may have dramatic impact on the choices taken (see Keren, 2010). For example, asking people to choose to accept, or instead reject, one of two candidates will focus them on either positive or negative reasons for their choice, leading to inconsistent decisions (Shafir, Simenson, & Tversky, 1993). So, again, intuitive and pragmatic processes rapidly provide the focus for reasoning, affecting its outcome.

Psychologists have been particular interested in decisions made under risk, which require estimation of probabilities and utilities associated with particular outcomes. Because these typically deviate from expected utility, much work has focused on judgments of subjective probabilities and how they are formed. In a famous series of papers published from the early 1970s onwards, Amos Tversky and Danny Kahneman proposed that such judgments are by reliance on *heuristics*, or shortcut rules of thumb. Because such heuristics were typically blamed for a number of observed cognitive biases, this became known as the "heuristics and biases" research program (Gilovich, Griffin, & Kahneman, 2002; Kahneman, Slovic, & Tversky, 1982). For example, reliance on a *representativeness*

heuristic was blamed for neglect of sample size (Kahneman & Tversky, 1972), neglect of base rates (Kahneman & Tversky, 1973) and the conjunction fallacy (Tversky & Kahneman, 1983). A judgment by representativeness is essentially one based on similarity of salient features. For example, when asked to judge the likelihood that a sample is drawn from a particular population, people tend to compare their mean scores, neglecting other statistically relevant features, such as sample size.

This research program is controversial and for our purposes also has an important ambiguity. The two issues are (a) are heuristics good or bad, and (b) are they based on explicit (Type 2) or implicit (Type 1) processes? Let me consider these separately. Kahneman and Tversky's work was widely cited as showing evidence of human irrationality, leading to a joint attack on their work and the early studies of Peter Wason (Cohen, 1981) from a rationalist perspective. Their work was also criticized by Gerd Gigerzner (e.g., Gigerenzer, 1991), who subsequently developed a research program of his own in which the adaptive value of heuristics was stressed (Gigerenzer, Todd, & the ABC Research Group, 1999) known as the "fast and frugal heuristics" program. A neutral view might be that heuristics are neither good nor bad. As originally pointed by Newell and Simon (1972), they are shortcut solutions to complex problems that may save enormous amounts of processing effort but that are not guaranteed to succeed. Thus we might see these two programs as focusing on cases where heuristics are useful, on one hand, and lead to biases, on the other.

The more pertinent question for our purposes is (b). Are heuristics Type 1 or Type 2? In the reasoning literature, the term "heuristic processes" is often used as a synonym for Type 1 or intuitive processing. On the other hand, *an* heuristic can be thought of as a simple but explicit rule that someone can follow deliberately. The early publications on heuristics were quite ambiguous in this respect. For example, the availability heuristic (Schwarz & Vaughn, 2002; Tversky & Kahneman, 1973) is defined as judging the probability of an event by the ease with which items come to mind. This could be intuitive: We might *feel* that events are probable because they easily activate our memory system. It could also be reflective: We might *reason* that events easily recalled must have occurred more frequently. Both Kahneman and Gigerenzer seem to agree that heuristics are Type 1. Kahneman and Frederick (2002) retrospectively adopted dual process theory (but see Kahneman & Tversky, 1982, for a forerunner), attributing biases to System 1 that may or may not be overridden by explicit System 2 reasoning. By the latter device, they explain in common with Stanovich (1999, 2010) why participants may sometimes avoid the dominant bias and find a normative solution. Gigerenzer is a well-known critic of dual-process theory (e.g., Gigerenzer & Regier, 1996), but his advocacy of reliance on "gut feelings" in judgment (Gigerenzer, 2007) nevertheless suggests that he sees heuristics as intuitive.

I am inclined to agree with Betsch (2008) that these literatures confuse two different kinds of judgment, which he calls intuitive and heuristic. Intuitive judgement is based on feelings of rightness or confidence and is therefore Type 1. Heuristic judgement, on the other hand, is based on simple but explicit rules and is therefore Type 2. Although in general Type 1 processing is quicker than Type 2, it is nonetheless a fallacy to believe that a process is necessarily Type 1 just because it is quick (Evans, in press). In common with Gigernzer, Gladwell (2005) adopted the jingle "less is more" to advocate the merits of intuitive judgment. But this phrase is ambiguous: What exactly is less? The time taken? The amount of information processed? The effort required? The first case that Gladwell discussed in his book is of an art expert who immediately "knows" that a statute is a fake but is unable to prove it for some time after. This is clearly a case of intuitive judgment. However, a later example of "less is more" is that of a hospital emergency room examining patients with chest pains. He shows how a hospital that replaced a complex series of diagnostic tests with just three simple questions achieved better results in trying to determine whether the patient should be treated as having a suspected heart attack. But this is clearly heuristic and not intuitive judgement. Explicit rule following is a Type 2 process however short and simple the rules may be.

One of the most widely investigated phenomena inspired by the heuristics and biases programme is often known as the "base rate fallacy." According to Bayes' theorem, which is a direct consequence of the probability calculus, posterior judgements about a hypothesis given some evidence should depend on two multiplicative factors: (a) the prior probability of the hypothesis and (b) the diagnosticity of the evidence, that is the extent to which it favors the hypothesis under consideration relative to rival hypotheses. For example, a positive medical test for a disease might be a hundred times more likely to occur in someone with the disease than without it. But if the base rate probability of having the disease in the population is 1 in 1,000, then it is still very likely to be false positive result. Early research by Kahneman and Tversky (1973) showed that people will focus strongly on the diagnostic information to the neglect of the base rate or prior probability, a finding that they attributed to the representativeness heuristic. Due to simplified and inaccurate citation, this led to myth that people ignore base rates, whereas the evidence shows that they underweight them (J. J. Koehler, 1996). However, it is generally agreed that even the *relative* neglect of base rates is an important cognitive bias affecting decision making in medicine and other real-world applications.

Correct Bayesian reasoning is really difficult to achieve for someone not trained in statistics. Even participants of higher cognitive ability neglect base rates unless pragmatic cues to their relevance are provided (Stanovich, 1999). You need both to somehow derive Bayes' theorem, which is not given by the experimenter, and to do the arithmetic necessary to combine base rates and diagnostic data. People are generally good at it only when probabilities are both presented as frequencies and arranged in "nested sets" (Barbey & Sloman, 2007; Cosmides & Tooby, 1996; Gigerenzer & Hoffrage, 1995; Hoffrage, Gigerenzer, Krauss, & Martigon, 2002). For example, if you are told that out of 1,000 people, 1 will have the disease and test positive but 50 healthy people will also test positive, then people have little difficulty in saying (correctly) that a person with a positive result is only about 2% likely to have the disease. If instead you are simply told that the false positive rate for a medical test is 1 in 20 and the base rate for the condition is 1 in 1,000, people most commonly say that a person with a positive test results is 95% likely to have the disease! (Cosmides & Tooby, 1996; Evans, Handley, Perham, Over, & Thompson, 2000). The nested sets presentation appears to cue a mental model that makes the logical relationships transparent and thus facilitates Type 2 reasoning.

Although the intuitive response to the complex version of this problem is typically based on the diagnostic information, there is still some (fallacious) reasoning involved. People must argue something like, "The test has a false positive rate of 1 in 20, which means it is right 95% of the time." Moreover, as has not often been noted in this literature, these studies also showed that a minority of participants give the answer 1 in 1,000, thus matching the *base rate*. What they are apparently unable to do is to integrate the two pieces of information. Clearly, the standard probability presentation of the Bayesian reasoning problem fails to meet a key criterion for the intervention of good reasoning: Participants must have the "mindware" for the task (Stanovich, 2010). Without the relevant education, no one carries Bayes' theorem around in their heads. But why do most people focus on the diagnostic rather than base rate information? My colleagues and I hypothesised that this was due to a bias in the normal format of the task. In many studies the diagnostic data are more intuitive because they are based on stereotypes and existing beliefs, whereas prior probabilities are conveyed only by explicit base rate information. We managed to devise a task in which this was switched around: Base rates were conveyed by genuine prior beliefs and diagnostic information by statistics (Evans, Handley, Over, & Perham, 2002). This shifted the balance of power and in one experiment led to base rates dominating the judgments. So we can see that a form of belief bias operates in this paradigm.

The correct conclusion to draw from research on Bayesian reasoning and base rate neglect is that this is not a problem that can be reliably solved by intuition, not because people are stupid or irrational but simply because they do not have the relevant rules available without training. Of course, it then follows that where good Bayesian reasoning is required for professional expertise, such training should be provided. This includes not only doctors but those in other professions who sometimes need statistical reasoning, for example, lawyers and judges who are required to interpret DNA evidence, which can easily result in fallacious reasoning in the untrained (Gigerenzer, 2002).

The judgment and decision literature has lagged behind the psychology of reasoning in the adoption of process-tracing measures such as response latencies, verbal protocols, and neural-imaging as well as the use of other dual-process manipulations such as working memory load, speeded tasks, and correlate with cognitive ability measures. For this reason, it is not easy to sort out what in these literatures is genuinely intuitive in the sense of being based on Type 1 processing. However, there is recent evidence that supports Kahneman and Frederick's (2002) proposal that heuristics and biases should be interpreted within a dual-process framework. Take the case of the conjunction fallacy, famously illustrated by the Linda problem (Tversky & Kahneman, 1983). Participants are given a brief description of Linda as follows:

> Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in antinuclear demonstrations.

When asked to rank a series of statements about Linda, the original study showed that, on average, participants rated "Linda is a bank teller and is active in the feminist movement" as significantly more likely than "Linda is a bank teller." However, this amounts to an assertion that $P(A\&B) > P(A)$, which cannot be true for any A and B by the laws of probability (and those of simple logic). On the face of it this seems a clear bias of intuitive judgment. On the description given you would not expect Linda to be a bank teller but you would expect her to be a feminist. When the feminist description is added, the statement *feels* more probable.

What has emerged in recent research is that performance on the Linda problem and a number of other decision problems is better among those of higher cognitive ability, but *only if a within participant design is used* (Stanovich & West, 2008). In the between-participant design, individuals simply rate the probability of one or the other statement, and the conjunction will be rated higher, regardless of cognitive ability.

Why should this be? Stanovich (2010) agreed with Kahneman and Frederick (2002) that finding the correct solution on problems like this requires "System 2" intervention. However, he has refined his earlier claims that such interventions are more likely to occur in those of higher ability (Stanovich, 1999). This is now viewed as only one of several requirements for default intuitions to be overridden by reasoning. One, as previously mentioned, is possession of the relevant "mindware." Another is the perceived need for reasoning, which may be influenced by a number of factors such as instructions given, feelings of confidence in the intuitive answer, and the rational thinking disposition of the individual. The point about the within-participant design is that individuals can see that they are wanting to assign a higher value to P(A&B) than to P(A), which may cue the need for reasoning and the detection of the inconsistency. This now makes it comparable to a belief bias experiment in which high-ability participants will inhibit beliefs but *only* if given strict deductive reasoning instructions of the kind described earlier (Evans, Handley, Neilens, Bacon, & Over, 2010). Hence, intervention on intuitions by reasoning requires both the cognitive capacity for the relevant reasoning and the awareness of the need for doing so.

If one assumes the perspective of the normative theory that is built into our research paradigms, then the psychology of reasoning appears to be unconnected with the study of judgment and decision making. The former assess people's ability to make certain deductions from assumed premises; the latter field investigates people's ability to choose actions by calculating and comparing their consequences. The psychological reality, however, is quite different. The two research fields have similar methods and similar findings. Researchers in both fields give participants problems to solve in the laboratory for which they lack any requisite training. Most people (consciously or not) rely on intuitive feelings about the answers and frequently get them wrong. A small number of exceptionally intelligent participants manage to reason out the answers from first principles, but even then they usually need help in the form of instructions or structural clues in the problem to get them on the right track. Perhaps the approach is fundamentally misconceived. One could argue that we will learn more about rationality by studying people's ability to apply rule-based reasoning that they have been taught, rather than by constantly studying the performance of naïve participants on novel problems.

Conclusions

The case for distinctively human powers of reasoning and reflective thinking that underlie the unique achievements of our species seems beyond doubt. However, in reviewing the case for the two minds hypothesis (Evans, 2010), I was struck over and over by the dominance of intuition in control of our behavior. The belief that we have in ourselves as conscious controlling agents does seem to be largely illusory. When we have a severe phobia, insight into the condition may not help us to escape it; if we have a gambling habit, our powers of reasoning merely serve to rationalize our behavior and confabulate erroneous theories of chance events. As I have shown in this article, when we study naïve reasoning and decision making in the laboratory, immediate, intuitive, and effort-free answers strongly compete with those based on reflective reasoning, often resulting in cognitive biases.

Although historically, philosophers and other scholars have stressed the importance of (logical) reasoning to achieve rationality, there has been a recent fashion for emphasising instead the value of reliance on intuition. From a dual-process perspective, neither view is correct. The evidence suggests that intuition is the dominant basis for real world decision making and is often effective; however, it also shows that reliance on intuition can be dangerous and that intervention with high-effort and explicit reasoning is often required, especially when problems have novel features. The problem with doing what feels right is that one has no reasoned basis for one's actions. This problem is greatly exacerbated by our tendency to reason after the event, confabulating justifications for our intuitions and constructing an illusion of conscious control.

Although the value of focusing on naïve participants and novel problems in the study of reasoning and decision making is debatable, this practice has taught us some important lessons. The first and most obvious is that people are not good at finding normative solutions to problems in logic, probability, and decision theory without special training. The massive evidence of cognitive biases on these tasks needs to be explained, for these are systematic and not random errors. Although some of these can be attributed to limits in working memory capacity and in reasoning strategy, many turn out to reflect reliance on intuition. Moreover, these intuitions are convincing to the individual—they feel right, they have confidence in them, and they frequently deter them from more effortful reasoning. Worse still, we can always find a rational sounding explanation for our actions.

Why is it that intuition so often dominates reasoning? I think there are two answers to this. The first is that intuitive feelings largely reflect experiential learning. In the real world, dealing with familiar environments, intuition will often serve us well. The second reason is to do with fundamental cognitive architecture of the human mind. Although intuitive processes operate rapidly, in parallel and with no effort, reflective reasoning is quite the opposite. Reflection requires

use of central working memory: There is only one such system—it has low capacity, requires high effort, and can be applied only to one task at a time. We are miserly with this cognitive resource because we have to be. The task of psychologists is to identity when intuition can safely be relied upon and when it cannot. From an educational viewpoint, we need not just to find ways of training people to reason well. We also need to find ways of making people aware of the situations in which such effortful processing needs to be applied.

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